Computer Science & DA

Warehousing

Lecture - 04

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Recap of Previous Lecture

Topic Datawarehouse Schema

Topic Star Schema

Topic Snowflake Schema

Topics to be Covered

- **Topic** Data Transformation
- **Topic** Data Cleaning
- **Topic** Data Smoothing

Topic : Data Transformation

1. Data Cleaning
2. Data Smoothing
3. Attribute Construction
4. Data Aggregation
5. Data Discretization
6. Data Generalization
7. Data Normalization

Topic : Data Cleaning

Removing or correcting errors, inconsistencies, and missing values in the data

Topic : Data Smoothing

Process to remove noise from the dataset using some algorithms

It allows for highlighting important features present in the dataset

Topic : Data Generalization

It converts low-level data attributes to high-level data attributes using concept hierarchy

Example:

Categorical attributes, such as house addresses, may be generalized to higher-level definitions, such as town or country

Topic : Data Normalization

Data normalization involves converting all data variables into a given range

- **Normalization Techniques:**
  1. Min-Max Normalization
  2. Z-score Normalization
  3. Decimal Scaling

Employees

Eid	Ename	age	Salary
1	Amit	39	25000000
2	Riya	40	35000000
3	Mahesh	36	1500500

Topic : Data Transformation

- **Advantages:**

1. Better Organization
2. Improved Data Quality
3. Perform Faster Queries
4. Better Data Management
5. More Use Out of Data

Topic : Data Transformation

- **Disadvantages:**
  1. Expensive
  2. Resource Extensive
  3. Data Loss
  4. Biased transformation
  5. Overfitting
  6. Can introduce problems due to lack of expertise and carelessness

Topic : Data Cleaning

Removing or correcting errors, inconsistencies, and missing values in the data.

Topic : Data Cleaning/cleansing/scrubbing

Data cleaning is the process of fixing or removing incorrect, corrupted, incorrectly formatted, duplicate, or incomplete data within a dataset.

Topic : Data Cleaning

Data cleaning is the process of fixing or removing incorrect, corrupted, incorrectly formatted, duplicate, or incomplete data within a dataset

1. Remove Duplicates
2. Fix structural errors
3. Handle missing data

Topic : Data Cleaning

Characteristics of clean data

- accuracy
- completeness
- consistency
- integrity
- timeliness
- uniformity
- validity

Topic : Data Cleaning

Benefits of data cleaning

- Improved decision-making
- More effective marketing and sales
- Better operational performance
- Increased use of data
- Reduced data costs

Topic : Data Cleaning

students

Name	id	Marks
Amit	S1	
Sunil	S2	

Challenges of data cleaning

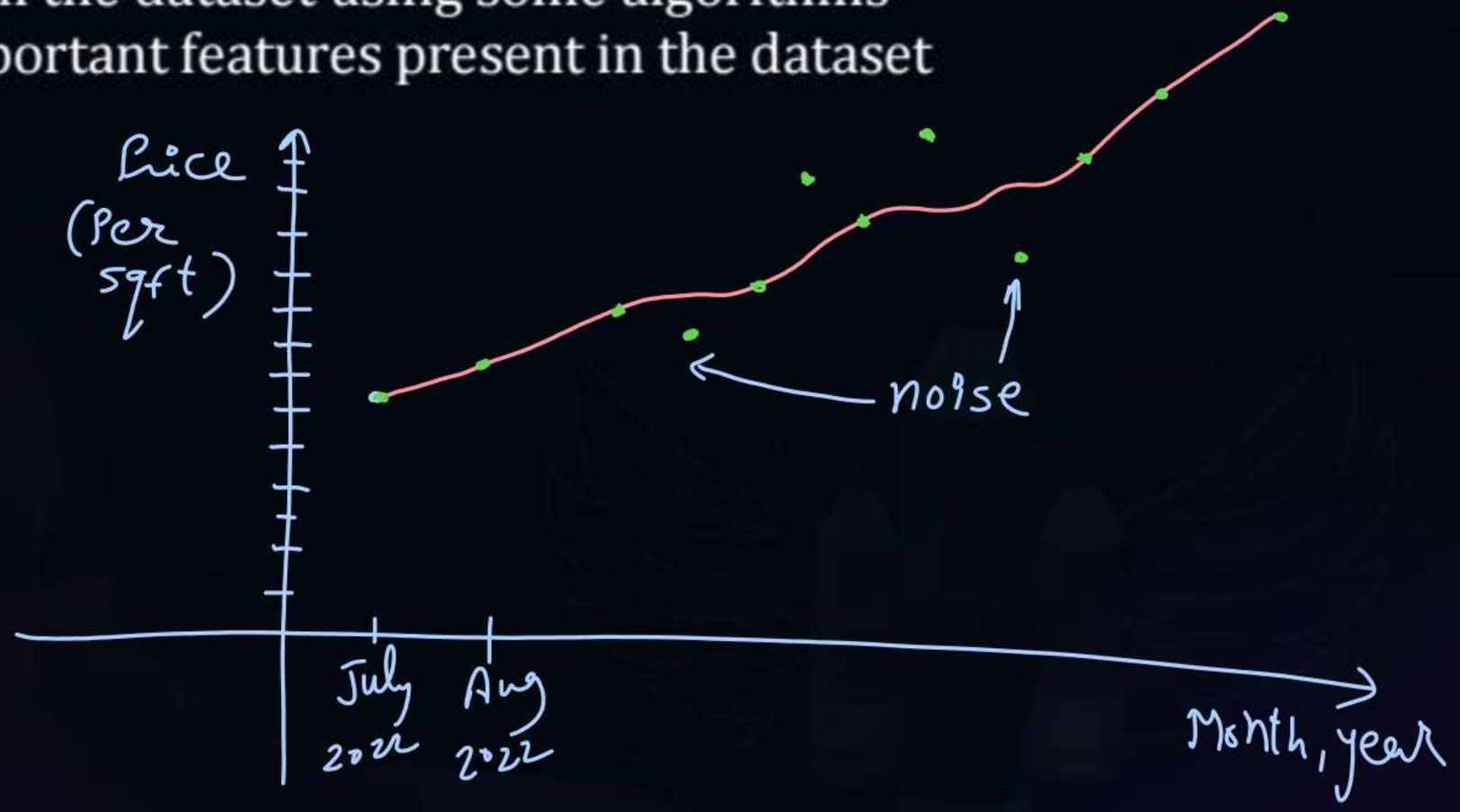
- Deciding how to resolve missing data values so they don't affect analytics applications
- Fixing inconsistent data in systems controlled by different business units
- Cleaning up data quality issues in big data systems that contain a mix of structured, semi-structured and unstructured data
- Getting sufficient resources and organizational support
- Dealing with data silos that complicate the data cleansing process

Topic : Data Smoothing

Process to remove noise from the dataset using some algorithms
It allows for highlighting important features present in the dataset

Bangalore

Month, year	Per sqft Price
July 2022	7000
Aug 2022	7200
Sep 2022	8000
⋮	
July 2024	15000



Topic : Data Smoothing

1. Binning
2. Moving Average
3. Random walk
4. Random method
5. Exponential smoothing

Topic : Binning / Data Discrete Binning / Data Bucketing

The original data values which fall into a given small interval, a bin, are replaced by a value representative of that interval, often a central value (mean or median). Binning is a generalization of rounding.

5, 9, 12, 13, 19

median

5, 9, 13, 19

$Avg = \frac{9+13}{2} = 11$

median

Topic : Binning

1. Smoothing by bin means
2. Smoothing by bin boundaries

Topic : Smoothing by Bin Means

→ replace each value of bin by mean

8, 16, 9, 15, 21, 21, 24, 30, 26, 27, 30, 34

1. Sort
2. Binning
3. Replace

8, 9, 15, 16, 21, 21, 24, 26, 27, 30, 30, 34

Assume,
Bin size = 4

$\boxed{8, 9, 15, 16}$, $\boxed{21, 21, 24, 26}$, $\boxed{27, 30, 30, 34}$

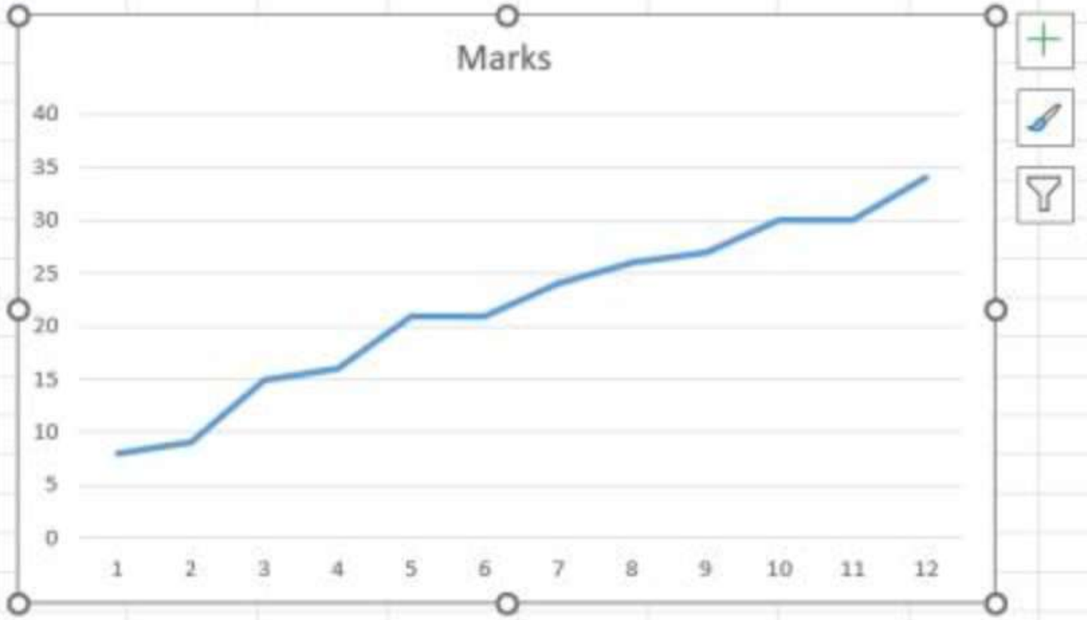
⇓ avg = 12 ⇓ avg = 23 avg = 30

12, 12, 12, 12 23, 23, 23, 23 30, 30, 30, 30

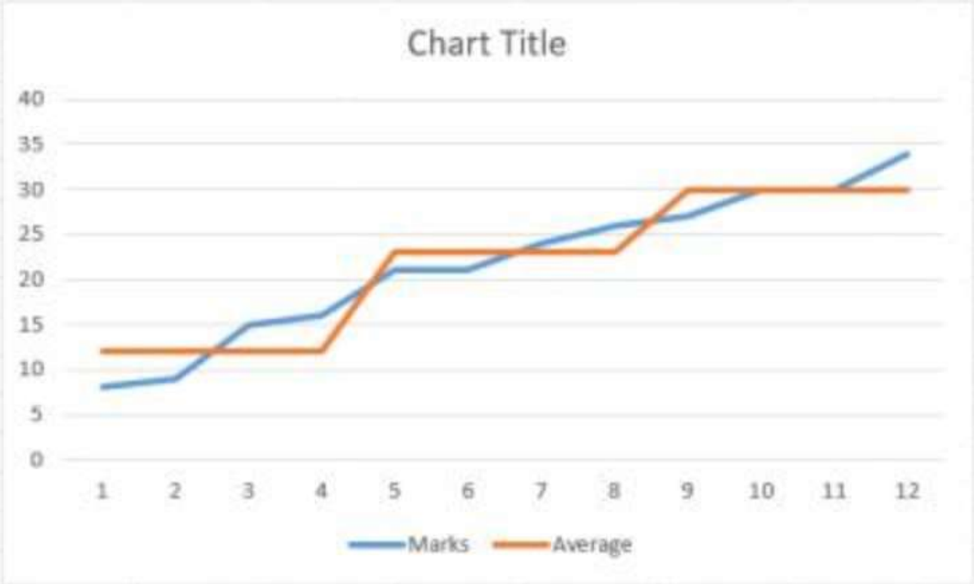
$$avg = \frac{8+9+15+16}{4}$$

$$= 12$$

Sequence	Marks
1	8
2	9
3	15
4	16
5	21
6	21
7	24
8	26
9	27
10	30
11	30
12	34



Sequence	Marks	Average
1	8	12
2	9	12
3	15	12
4	16	12
5	21	23
6	21	23
7	24	23
8	26	23
9	27	30
10	30	30
11	30	30
12	34	30



Topic : Smoothing by Bin Boundary

↳ replace internal values by nearest boundary value

8, 16, 9, 15, 21, 21, 24, 30, 26, 27, 30, 34

8, 9, 15, 16, | 21, 21, 24, 26, | 27, 30, 30, 34

8, 8, 16, 16 21, 21, 26, 26 27, 27, 27 34

Tie breaker:-

No standard given, hence it will be mentioned in questn.

Topic : Smoothing by Binning

Homework

50, 6, 2, 48, 42, 34, 29, 25, 15, 45, 15, 45, 49, 12, 25, 2

Make bins of 4 values

1. Smoothing by bin mean
2. Smoothing by bin boundary

2 mins Summary

Topic Data Transformation

Topic Data Cleaning

Topic Data Smoothing

Mon - Friday

7:30 am

Happy Learning

THANK - YOU